

From Inside Game to Outside Game

Tracing the NBA's Three-Point Revolution, 2000-01 to 2025-26

DSAI 691 Relational Databases, Group Project Phase 2: Populate PostgreSQL and Install Metabase

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Deliverable	Where
Database and tables created	create_and_load.sql Part A and B; section 1
Data imported with COPY	create_and_load.sql Part B; section 1.1
Understanding the data with SQL	create_and_load.sql Part C; section 2
Metabase installed on Docker	sections 3.1 and 3.2
Metabase connected to PostgreSQL	sections 3.3 and 3.4

Data source: stats.nba.com LeagueGameLog via the nba_api Python client, 26 seasons, 62,508 team-game rows after cleaning. Raw JSON is exported to CSV by cleaning/export_csv.py and loaded with COPY; the three CSV files are submitted in data/csv/.

PART 1. Database and tables in PostgreSQL

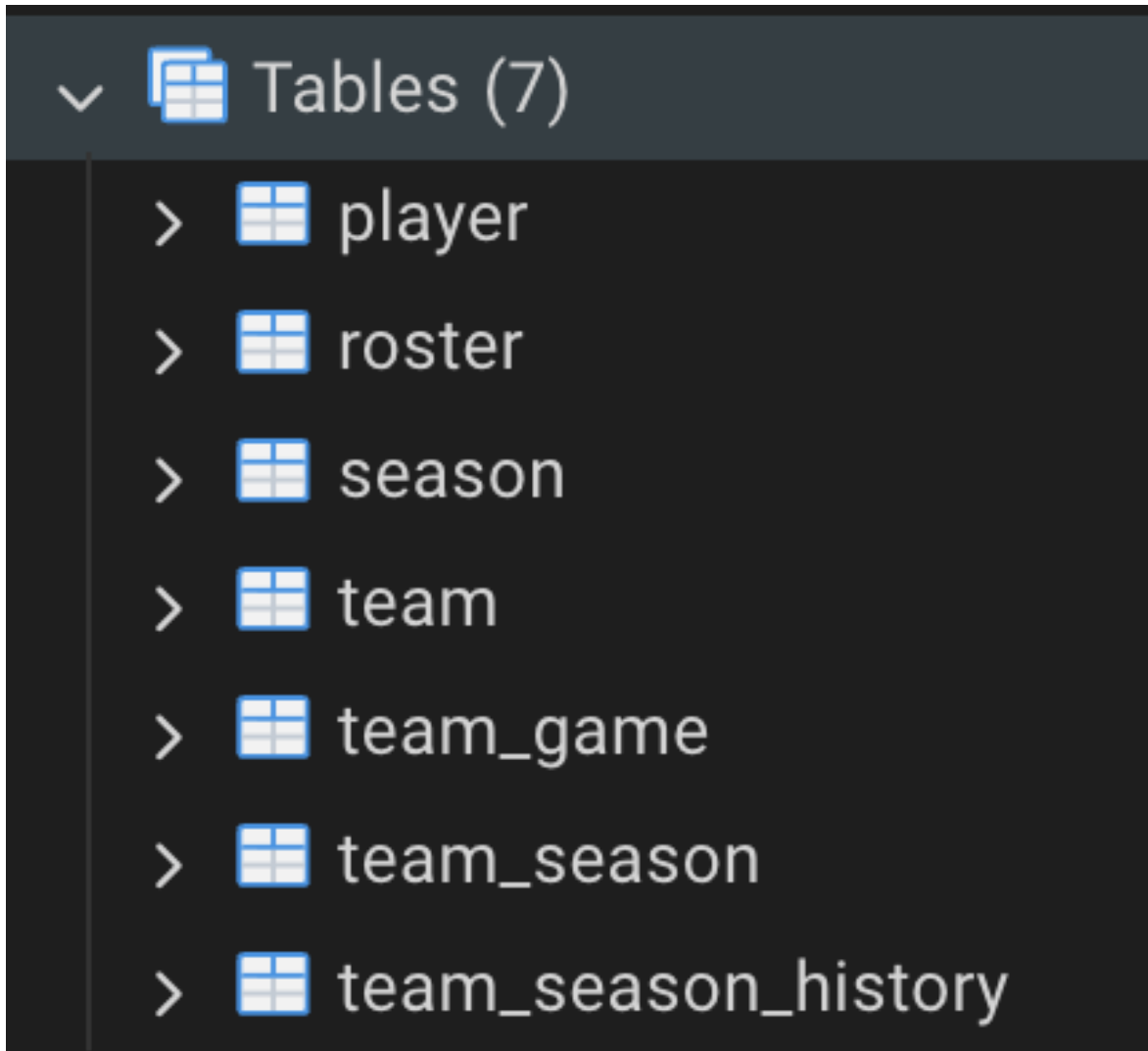
1.1 Data loaded: row counts

Result of the final validation query in Part B of create_and_load.sql, run in pgAdmin. All five populated tables match expectations. team_season and team_season_history have 776 rows rather than 780 because the league had 29 teams until the Charlotte Bobcats joined in 2004-05. problem_games = 0 confirms every game has exactly two team rows, one winner and at most one home team. player and roster are created with their keys but are loaded in Phase 3.

	table_name text	row_count bigint
1	roster	0
2	player	0
3	season	26
4	team	30
5	team_season_hist...	776
6	team_season	776
7	team_game	62508
8	problem_games	0

1.2 The seven tables in pgAdmin

Schema nba_three_point_revolution.public after running the script.



PART 2. Understanding the data with SQL

2.1 The three-point curve (query C1)

Share of all field-goal attempts taken from three, by season. It rises from 17.0% in 2000-01 to 41.5% in 2025-26, with the steepest climb after 2014-15.

	season [PK] text 	era_label text 	pct_shots_from_three numeric 	threes_attempted_per_team_game numeric 
1	2000-01	pre-revolution	17.0	13.7
2	2001-02	pre-revolution	18.1	14.7
3	2002-03	pre-revolution	18.2	14.7
4	2003-04	pre-revolution	18.7	14.9
5	2004-05	pre-revolution	19.6	15.8
6	2005-06	pre-revolution	20.2	16.0
7	2006-07	pre-revolution	21.3	16.9
8	2007-08	pre-revolution	22.2	18.1
9	2008-09	pre-revolution	22.4	18.1
10	2009-10	acceleration	22.2	18.1
11	2010-11	acceleration	22.2	18.0
12	2011-12	acceleration	22.6	18.4
13	2012-13	acceleration	24.3	20.0
14	2013-14	acceleration	25.9	21.5
15	2014-15	acceleration	26.8	22.4
16	2015-16	post-revoluti...	28.5	24.1
17	2016-17	post-revoluti...	31.6	27.0
18	2017-18	post-revoluti...	33.7	29.0
19	2018-19	post-revoluti...	35.9	32.0
20	2019-20	post-revoluti...	38.4	34.1
Total rows: 26		Query complete 00:00:00.096		

2.2 Volume changed, accuracy did not (query C2)

League three-point percentage stays between 35.5% and 35.9% across all three eras while the share of shots from three nearly doubles. Teams did not get better at threes; they chose to take many more.

	era_label text	first_season text	last_season text	three_point_pct numeric	pct_shots_from_three numeric
1	pre-revolution	2000-01	2008-09	35.7	19.8
2	acceleration	2009-10	2014-15	35.5	24.1
3	post-revoluti...	2015-16	2025-26	35.9	37.2

2.3 Who moved first (query C3)

Top three teams by three-point rate each season (window function RANK). Boston led the early 2000s; Phoenix led from 2004-05. Relocated franchises appear under their current name because the NBA keeps one team_id per franchise.

	season text	season_rank bigint	team_name text	pct_shots_from_three numeric
1	2000-01	1	Boston Celtics	25.2
2	2000-01	2	Miami Heat	22.1
3	2000-01	3	Milwaukee Bucks	21.8
4	2001-02	1	Boston Celtics	28.9
5	2001-02	2	Orlando Magic	24.1
6	2001-02	3	Milwaukee Bucks	24.1
7	2002-03	1	Boston Celtics	33.1
8	2002-03	2	Dallas Mavericks	23.9
9	2002-03	3	Detroit Pistons	23.7
10	2003-04	1	Oklahoma City Thu...	29.4
11	2003-04	2	New Orleans Pelic...	25.2
12	2003-04	3	Boston Celtics	24.9
13	2004-05	1	Phoenix Suns	28.9
14	2004-05	2	Oklahoma City Thu...	28.1
15	2004-05	3	Los Angeles Lakers	27.3
16	2005-06	1	Phoenix Suns	29.3
17	2005-06	2	Golden State Warri...	27.0
Total rows: 78		Query complete 00:00:00.122		

2.4 Three-point volume and accuracy against winning (query C4)

Per-season correlation between a team's three-point attempts per game and its win percentage, and between three-point accuracy and win percentage. Accuracy correlates with winning more strongly than volume in 25 of the 26 seasons.

	season text	corr_3pa_volume_vs_win_pct numeric	corr_3p_accuracy_vs_win_pct numeric
1	2000-01	0.34	0.48
2	2001-02	0.33	0.43
3	2002-03	0.25	0.50
4	2003-04	-0.05	0.30
5	2004-05	0.39	0.55
6	2005-06	0.28	0.27
7	2006-07	0.33	0.42
8	2007-08	0.38	0.54
9	2008-09	0.26	0.48
10	2009-10	0.26	0.53
11	2010-11	0.25	0.42
12	2011-12	0.14	0.57
13	2012-13	0.27	0.53
14	2013-14	0.25	0.53
15	2014-15	0.47	0.69
16	2015-16	0.25	0.64
17	2016-17	0.28	0.64
Total rows: 26		Query complete 00:00:00.053	

2.5 Best records of the last decade (query C5)

The 15 best regular-season records since 2015-16 with each team's three-point profile, joining team_season_history, team and team_season.

	season text 🔒	team_name text 🔒	wins integer 🔒	losses integer 🔒	win_pct numeric 🔒	fg3a_per_game numeric 🔒	fg3_pct numeric 🔒
1	2015-16	Golden State Warriors	73	9	0.890	31.61	0.416
2	2024-25	Oklahoma City Thun...	68	14	0.829	38.83	0.374
3	2015-16	San Antonio Spurs	67	15	0.817	18.51	0.375
4	2016-17	Golden State Warriors	67	15	0.817	31.24	0.383
5	2017-18	Houston Rockets	65	17	0.793	42.32	0.362
6	2021-22	Phoenix Suns	64	18	0.780	31.90	0.364
7	2024-25	Cleveland Cavaliers	64	18	0.780	41.48	0.383
8	2025-26	Oklahoma City Thun...	64	18	0.780	37.94	0.365
9	2023-24	Boston Celtics	64	18	0.780	42.46	0.388
10	2019-20	Milwaukee Bucks	56	17	0.767	38.90	0.355
11	2025-26	San Antonio Spurs	62	20	0.756	37.90	0.359
12	2024-25	Boston Celtics	61	21	0.744	48.23	0.368
13	2016-17	San Antonio Spurs	61	21	0.744	23.50	0.391
14	2019-20	Toronto Raptors	53	19	0.736	36.99	0.374
15	2018-19	Milwaukee Bucks	60	22	0.732	38.22	0.353

2.6 Home-court advantage over time (query C6)

Home win percentage by season. 2011-12 has 990 games (lockout) and 2012-13 has 1,229 because one game was cancelled after the Boston Marathon bombing and dropped during cleaning.

	season text	home_win_pct numeric	home_games bigint
1	2000-01	59.8	1189
2	2001-02	59.1	1189
3	2002-03	62.8	1189
4	2003-04	61.4	1189
5	2004-05	60.5	1230
6	2005-06	60.3	1230
7	2006-07	59.1	1230
8	2007-08	60.1	1230
9	2008-09	60.8	1230
10	2009-10	59.4	1230
11	2010-11	60.4	1230
12	2011-12	58.6	990
13	2012-13	61.2	1229
14	2013-14	58.0	1230
15	2014-15	57.5	1230
16	2015-16	58.9	1230
17	2016-17	58.4	1230
Total rows: 26		Query complete 00:00:00.159	

PART 3. Metabase

3.1 Metabase and PostgreSQL running as Docker containers

Docker Desktop, compose project `dsai691_nba_analytics_dashboard`, defined in `docker-compose.yml`. Three containers are running: Metabase (`nba_metabase`, image `metabase/metabase`, port 3000), PostgreSQL 16 (`nba_postgres`, image `postgres:16`, container port 5432 exposed to the host on 5433) and pgAdmin (`nba_pgadmin`, image `dpape/pgadmin4`, port 5050). The terminal pane at the bottom shows the same three containers in `docker ps` with status Up. Metabase connects to Postgres over the Docker network as host `db`, port 5432 (section 3.3).

The screenshot shows the Docker Desktop interface for the project `dsai691_nba_analytics_dashboard`. The left sidebar contains navigation options: Gordon, Containers, Images, Logs, Volumes, Kubernetes, Builds, Models, MCP Toolkit (BETA), Docker Hub, and Extensions. The main area displays the project details and a list of running containers:

- metabase** (image `metabase/metabase`, port 3000)
- db** (image `postgres:16`, port 5433)
- pgadmin** (image `dpape/pgadmin4`, port 5050)

The logs pane shows the following messages for the `metabase` container:

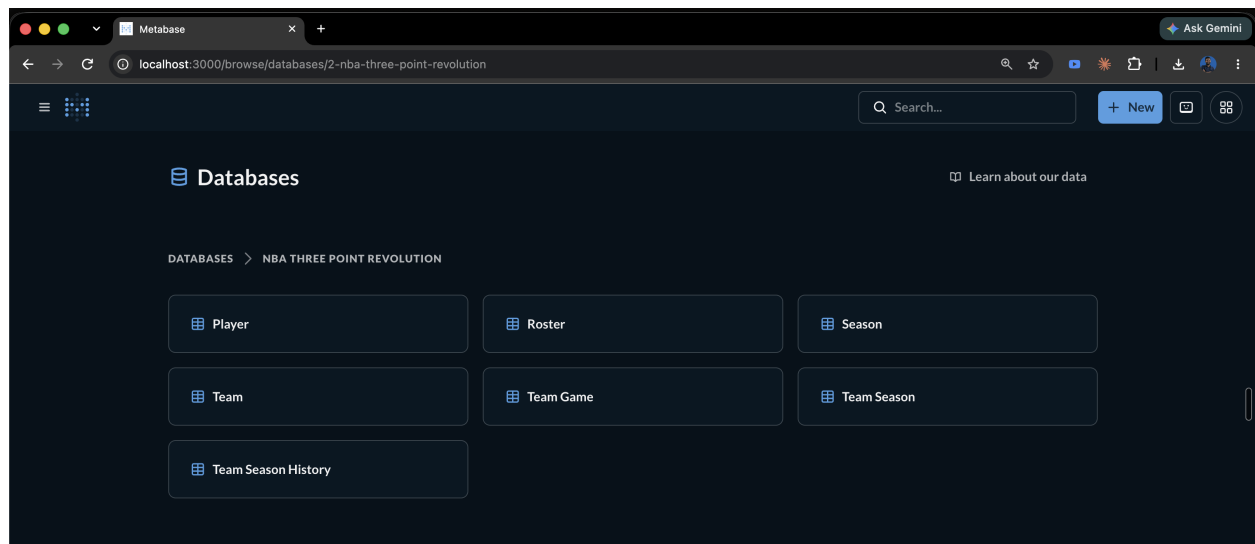
```
2026-09-25 14:01:25,030 INFO metabot.example-question-generator :: Generating example questions
natively (:table-count 1, :metric-count 2) {mb-quartz-job-type=SuggestedPromptsGenerator}
2026-09-25 14:01:25,041 INFO metabot.self :: Calling LLM (structured) (:provider anthropic,
:model claude-sonnet-4-6, :msg-count 1, :ai-proxy? false)
2026-09-25 14:01:25,056 WARN metabot.example-question-generator :: Example question generation
failed for one item: No Anthropic API key is set
2026-09-25 14:01:25,064 INFO metabot.self :: Calling LLM (structured) (:provider anthropic,
:model claude-sonnet-4-6, :msg-count 1, :ai-proxy? false)
2026-09-25 14:01:25,064 INFO metabot.self :: Calling LLM (structured) (:provider anthropic,
:model claude-sonnet-4-6, :msg-count 1, :ai-proxy? false)
2026-09-25 14:01:25,066 WARN metabot.example-question-generator :: Example question generation
failed for one item: No Anthropic API key is set
2026-09-25 14:01:25,068 ERROR task.suggested-prompts-generator :: Suggested prompts generation
failed: No Anthropic API key is set {mb-quartz-job-type=SuggestedPromptsGenerator}
```

The terminal pane shows the output of the `docker ps` command:

```
(base) yashtiwari@Yashs-MacBook-Pro ~ % docker ps
CONTAINER ID   IMAGE                     COMMAND                  CREATED    STATUS    PORTS
33796b318a40   metabase/metabase:latest  "/app/run_metabase.sh"  47 hours  Up 4 minutes    0.0.0.0:3000->3000/tcp, [::]
a035792d9407   postgres:16              "docker-entrypoint.s..."  2 days ago Up 4 minutes (healthy)  0.0.0.0:5433->5432/tcp, [::]
b358e94f02c5   dpape/pgadmin4:latest    "/entrypoint.sh"        6 days ago Up 4 minutes    0.0.0.0:5050->80/tcp, [::]:5050->80/tcp
(base) yashtiwari@Yashs-MacBook-Pro ~ %
```

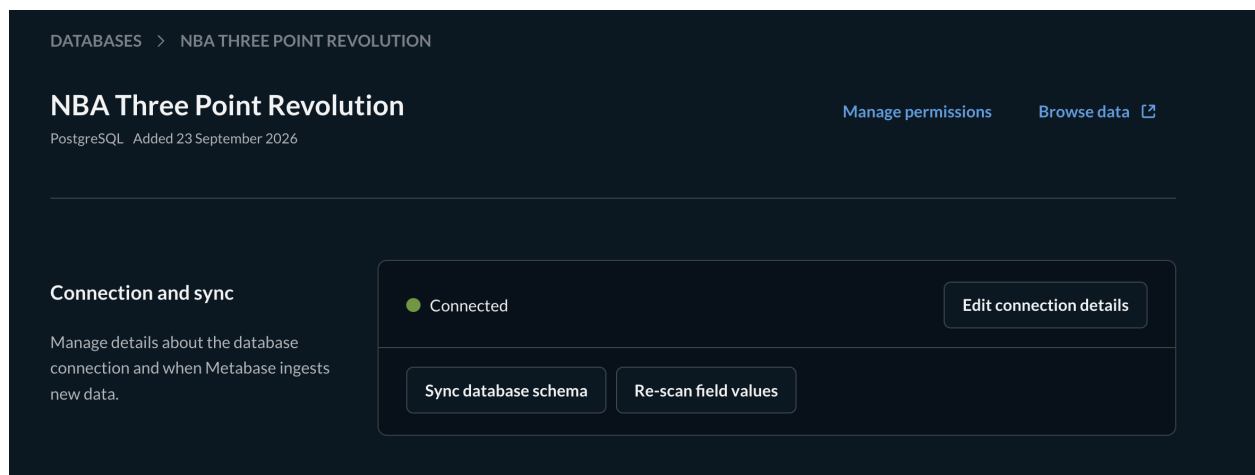
3.2 Metabase running in Docker, with our tables

Metabase served by the nba_metabase Docker container at localhost:3000 (address bar). Browse, Databases, NBA Three Point Revolution lists all seven tables synced from PostgreSQL.



3.3 Metabase connected to PostgreSQL

Admin settings, Databases: the NBA Three Point Revolution PostgreSQL database shows status Connected. Metabase reaches Postgres over the Docker network at host db, port 5432 (host port 5433).



3.4 A question built on the data

Native SQL question in Metabase against team_game, shown as a line chart: the share of shots from three rises from 17.0% in 2000-01 to a peak of 42.1% in 2024-25.

